

Open Mouth Is All You Need

Jaden Zhou

Khoury College of Computer Sciences

Northeastern University

zhou.jad@northeastern.edu

Abstract—Automatically detecting when a speaker is talking in monocular video is useful for lecture analytics, meeting summarization, and engagement monitoring. Most existing approaches rely on audio or deep visual models, obscuring which geometric cues actually drive performance. We investigate which hand-crafted facial features are stable and informative under natural appearance variation—lighting, head pose, and camera motion—using a single-subject corpus of 30 lecture recordings. A pipeline based on MediaPipe face landmarking extracts eight features per frame: mouth openness, eye openness, head-pose yaw and pitch (via PnP), frame-difference motion statistics, ambient brightness, and landmark detection confidence. Logistic regression and SVM-RBF classifiers are trained on one recording and evaluated on twenty held-out recordings without adaptation. An ablation study reveals that mouth openness alone achieves $F1=0.883$, matching the full eight-feature model ($F1=0.881$), while removing mouth openness drops performance to $F1=0.705$. Cross-video generalization across seventeen evaluable test recordings yields $F1=0.750\text{--}0.950$ (mean 0.848); three recordings are excluded due to degenerate class distributions or severe landmark occlusion and are analyzed separately. Pose and motion condition analysis reveals variable sensitivity across sessions, with larger frontal-vs.-angled gaps in recordings featuring extended head-down postures. A fine-tuning experiment confirms that adding up to 100 labeled target-domain frames provides no consistent improvement, further supporting the robustness of lip aperture as a cross-domain feature.

Index Terms—speaking detection, facial landmarks, MediaPipe, head pose, appearance variation, logistic regression, SVM

I. INTRODUCTION

Automated analysis of lecture and meeting recordings enables a range of downstream applications: identifying active speakers for indexing, estimating engagement from participation patterns, and generating structured transcripts aligned with visual activity. A fundamental sub-task is *frame-level speaking detection*—determining, for each video frame, whether the visible subject is currently producing speech.

Real-world recordings introduce appearance variation that can degrade feature reliability. Head pose changes as the speaker gestures or turns to write on a board. Ambient lighting shifts between windows, projectors, and overhead sources. Camera motion, zoom changes, and subject movement alter the pixel statistics of every frame. Prior work addressing talking and lip-movement detection has largely relied on deep convolutional or recurrent architectures [1], [2], which achieve high performance but provide little insight into which facial geometry actually carries the signal.

We take a *feature analysis* perspective: rather than maximizing accuracy with a black-box model, we ask which hand-

crafted geometric and photometric features are individually informative, which are redundant, and how stable each is under controlled appearance variation. Our experimental design uses a strict video-level train/test split across 30 recordings of a single subject, enabling direct measurement of cross-recording generalization without the confound of subject identity.

Contributions.

- 1) An ablation study across nine feature subsets demonstrating that mouth openness is both necessary and sufficient for speaking detection at this task difficulty, with no measurable gain from adding pose, motion, or brightness features.
- 2) A cross-video generalization study on twenty held-out recordings spanning the full semester, with condition-stratified evaluation (frontal vs. angled pose; low vs. high motion) identifying when and why performance degrades.
- 3) A fine-tuning experiment showing that adding small labeled target-domain samples (25–100 frames) provides no consistent benefit, confirming that mouth openness generalizes without adaptation.
- 4) A fully reproducible CPU-only pipeline using MediaPipe [5] and scikit-learn, with annotated ground truth and released code.

II. RELATED WORK

Lip reading and talking-face detection. Chung and Zisserman [1] trained a CNN on lip crops to synchronize audio and video, demonstrating that visual lip motion provides strong signal for speech activity. Chung et al. [2] extended this to sentence-level lip reading from frontal video, showing that deep networks can extract rich articulatory features—but requiring large annotated corpora and GPU-scale training. Our work differs by studying *which* geometric features drive performance in a resource-constrained, single-subject setting rather than maximizing accuracy with deep models.

Active speaker detection. Roth et al. [3] released the AVA-ActiveSpeaker dataset and benchmarked audio-visual models for detecting which face in a multi-speaker scene is talking. Their best systems fuse audio with visual lip features, underscoring the primacy of mouth region information—consistent with our ablation finding that mouth openness dominates. We focus on the visual-only case with a single subject, isolating feature geometry from audio and multi-face disambiguation.

Head pose estimation. Ruiz et al. [4] proposed a CNN-based head-pose estimator trained on synthetically augmented

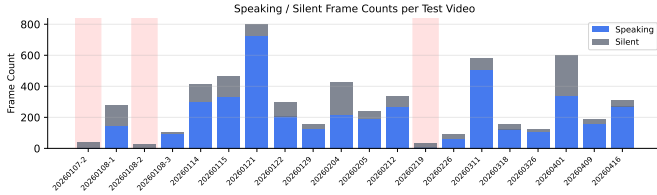


Fig. 1: Speaking and silent frame counts per test video (sorted chronologically). Red-shaded bars mark the three sessions excluded from aggregate statistics: 20260107-2 (2 speaking frames), 20260108-2 (0 speaking frames), and 20260219 (mouth frequently occluded).

data, achieving fine-grained yaw/pitch/roll without landmark supervision. We adopt the classical PnP approach using MediaPipe landmarks as a lightweight, interpretable alternative suitable for CPU-only deployment. Our generalization results confirm that pose variation up to the range present in lecture recordings does not significantly degrade mouth-openness-based speaking detection.

Face landmarking. Kartynnik et al. [5] introduced the MediaPipe Face Mesh model, producing 468 3D landmarks in real time on mobile hardware. We use this model as our feature extraction backbone, leveraging its landmark indices for mouth, eye, and pose computation.

III. METHODS

A. Dataset and Annotation

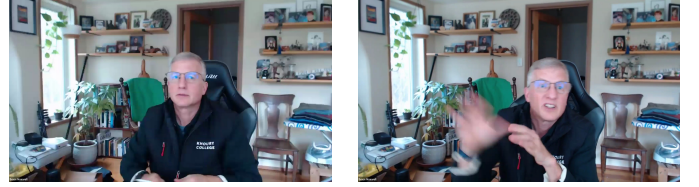
Our corpus consists of 30 lecture recordings of a single subject captured with a fixed webcam at standard definition. One recording (video 20260107-1, approximately 45 minutes) serves as the training split; all twenty remaining recordings spanning the full semester serve as the test split. Frames are extracted at a 10-frame interval (approximately 3 fps from a 30 fps source), yielding 1,179 labeled training frames and 5,656 labeled test frames across the twenty test videos. Two test recordings (20260107-2, 20260108-2) have zero or near-zero speaking instances and are excluded from aggregate metrics; one recording (20260219) is flagged as an outlier due to extended head-down posture (see Section V).

Ground-truth labels are binary: *speaking* (1) or *silent* (0). Labeling criterion is lip tension and movement continuity rather than instantaneous mouth aperture to avoid a circular dependency with the primary feature. Ambiguous frames (laughing, occluded face, mid-intake breath) are excluded from evaluation.

B. Feature Extraction

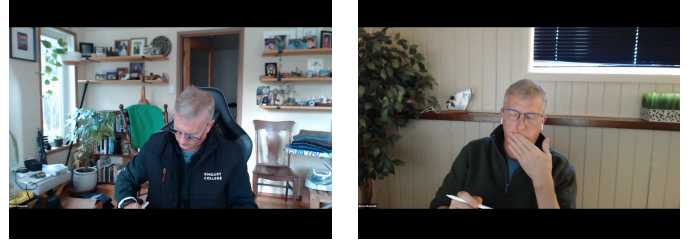
We extract eight features per frame using MediaPipe Face Landmarker (468 3D landmarks, minimum detection confidence 0.5):

- **Mouth openess:** Euclidean distance between landmarks #13 (upper inner lip) and #14 (lower inner lip), scaled by frame height. A higher value indicates a more open mouth.



(a) Frontal, silent (baseline)

(b) Speaking with hand gesture



(c) Head-down posture (mouth occluded)

(d) Hand covering mouth (occlusion)

Fig. 2: Representative frames illustrating appearance variation in the corpus. (a) Clean frontal baseline. (b) Active speech with gestural motion blur. (c) Head-down writing posture—the primary failure mode in 20260219, where the mouth exits the camera’s field of view. (d) Hand occlusion of the mouth region, a confound for landmark-based mouth openess.

- **Eye openess:** Average of left and right eyelid spans (landmarks #386/374 and #159/145 respectively), same normalization.
- **Yaw and pitch:** Head pose estimated via the Perspective-n-Point (PnP) algorithm [6] using six landmark-to-3D correspondences (nose tip, chin, eye corners, mouth corners). Yaw encodes left-right rotation; pitch encodes up-down tilt, both in degrees.
- **motion_mean, motion_var:** Mean and variance of pixel-wise absolute frame differences between consecutive grayscale frames, measuring camera and subject motion.
- **Brightness:** Mean grayscale intensity of the full frame (0–255), as a proxy for ambient lighting.
- **Landmark confidence:** Binary indicator (1.0 if landmarks detected, 0.0 otherwise).

Frames with missing landmark detections receive NaN for landmark-dependent features; such frames are excluded from all analyses via `dropna`.

C. Variation Conditions

To enable condition-stratified analysis, frames are assigned discrete appearance conditions using thresholds calibrated on the training distribution:

- **Pose:** frontal ($|\text{yaw}| < 15^\circ$) vs. angled ($|\text{yaw}| \geq 15^\circ$). The 15° boundary is standard in the pose literature and splits the training data approximately 38/62 frontal/angled.
- **Lighting:** high (brightness > 115) vs. low (≤ 115 , the 25th percentile of training brightness).
- **Motion:** low (motion_mean < 5.3) vs. high (≥ 5.3 , the training median, giving a near 50/50 split).

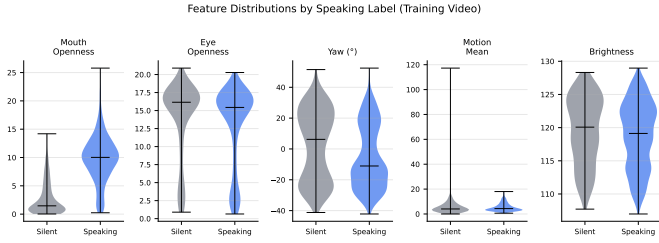


Fig. 3: Feature distributions by speaking label on the training video (violin plots; horizontal line = median). Mouth openness shows the clearest class separation. The remaining features have substantially overlapping distributions.

Absolute thresholds (rather than per-video relative thresholds) are used so that condition labels carry consistent real-world meaning across all test recordings.

D. Classifiers

Both classifiers operate on z -score normalized features (StandardScaler fit on training data only):

- **Logistic Regression** (L2 regularization, $C = 1$, `max_iter=1000`, `class_weight=balanced`).
- **SVM-RBF** (RBF kernel, `class_weight=balanced`).

Balanced class weighting compensates for the moderate speaking/silent imbalance present in several test recordings. Both models are trained once on all non-NaN training frames and applied without adaptation to each test video.

E. Evaluation Protocol

Experiment 1 (Ablation) uses five-fold stratified cross-validation on the training video to estimate within-distribution feature importance across nine feature subsets (Table I, Fig. 4). The primary metric is F1 score (harmonic mean of precision and recall), which is more informative than accuracy under class imbalance.

Experiment 2 (Generalization) evaluates the frozen model trained on 20260107-1 on each of the twenty test videos independently. Per-video F1, accuracy, and pose-condition (frontal vs. angled) F1 are reported (Table II, Fig. 5).

Experiment 3 (Fine-tuning) tests whether augmenting the training set with a small stratified sample of labeled frames from the target video recovers performance. Evaluated on five representative test videos with $n \in \{25, 50, 100\}$ added frames (Table III).

IV. EXPERIMENTS AND RESULTS

A. Experiment 1: Ablation Study

Fig. 3 shows the distribution of five features split by speaking/silent label on the training video. Mouth openness is the only feature with a clear separation between classes; the remaining features have substantially overlapping distributions, foreshadowing the ablation results.

TABLE I: Ablation Study — 5-fold CV F1 on Training Video

Feature Subset	LogReg F1	SVM-RBF F1
Eye only	0.444 ± 0.029	0.605 ± 0.031
Motion only	0.488 ± 0.023	0.693 ± 0.026
Brightness only	0.587 ± 0.028	0.746 ± 0.010
No mouth (7 feat.)	0.705 ± 0.030	0.697 ± 0.060
Pose only	0.717 ± 0.026	0.730 ± 0.051
All features (8)	0.881 ± 0.009	0.885 ± 0.012
No motion (6 feat.)	0.883 ± 0.008	0.883 ± 0.012
Mouth only	0.883 ± 0.009	0.879 ± 0.012
No pose (6 feat.)	0.885 ± 0.003	0.880 ± 0.011

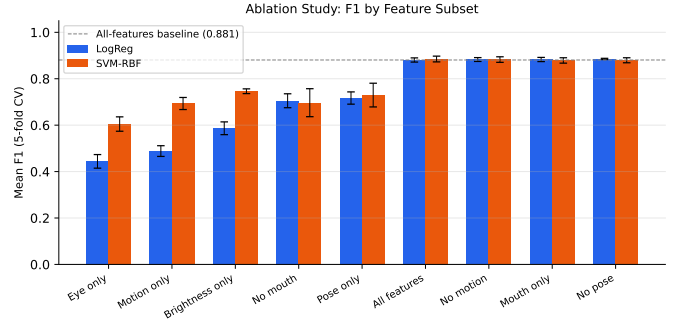


Fig. 4: Ablation study F1 by feature subset (5-fold CV, training video). Error bars show ± 1 standard deviation. Dashed line marks the all-features LogReg baseline (0.881).

Table I and Fig. 4 report mean F1 (± 1 std) from five-fold cross-validation on the training video for nine feature subsets and two classifiers.

The results reveal a clear bimodal structure. Individual features—eye openness (0.444), motion (0.488), brightness (0.587), and pose (0.717)—all fall well below the mouth-containing configurations. Notably, SVM-RBF narrows the gap for some single features (e.g., brightness 0.746, motion 0.693), suggesting that nonlinear boundaries capture weak signal; logistic regression does not. Any combination retaining mouth openness performs comparably: removing motion (0.883), removing pose (0.885), or using all eight features (0.881) all yield F1 statistically indistinguishable from mouth openness alone (0.883). Critically, removing mouth openness while retaining all seven other features drops LogReg F1 to 0.705—an 18-point absolute decrease.

Finding 1: Mouth openness is both necessary and sufficient for speaking detection in this corpus. No other single feature approaches its predictive power, and no combination of other features recovers comparable performance.

B. Experiment 2: Cross-Video Generalization

Table II and Fig. 5 show per-video results across all twenty test recordings. The model, trained on a single 45-minute recording, achieves $F1 = 0.750\text{--}0.950$ (mean 0.848) on the 17 evaluable recordings without any adaptation.

Pose sensitivity. The frontal vs. angled F1 gap (Δ_p) averages 0.074 across valid videos, with a wide range (0.000–0.262). Ten of seventeen videos show frontal F1 exceed-

TABLE II: Generalization — LogReg on All Twenty Test Recordings. $\Delta_p = |\text{frontal F1} - \text{angled F1}|$. Mean row excludes the three marked recordings.

Video	F1	Acc	Frontal	Angled	Δ_p
20260107-2 [†]	0.400	0.833	0.667	0.286	0.381
20260108-1	0.826	0.782	0.806	0.837	0.031
20260108-2 [‡]	0.000	0.115	—	—	—
20260108-3	0.945	0.896	0.936	0.967	0.031
20260114	0.950	0.924	0.956	0.929	0.027
20260115	0.793	0.683	0.819	0.672	0.147
20260121	0.941	0.895	0.928	0.946	0.018
20260122	0.899	0.846	0.874	0.909	0.035
20260129	0.863	0.773	0.937	0.791	0.146
20260204	0.912	0.906	0.897	0.916	0.019
20260205	0.770	0.674	0.787	0.698	0.089
20260212	0.752	0.664	0.743	0.761	0.018
20260219*	0.316	0.639	0.667	0.250	0.417
20260226	0.842	0.800	0.902	0.794	0.108
20260311	0.798	0.691	0.875	0.765	0.110
20260318	0.816	0.734	0.879	0.760	0.119
20260326	0.750	0.645	0.872	0.610	0.262
20260401	0.880	0.848	0.880	0.880	0.000
20260409	0.777	0.674	0.753	0.789	0.036
20260416	0.893	0.828	0.919	0.861	0.058
Mean (17)	0.848	0.780	0.869	0.817	0.074

[†]2 speaking instances; F1 unreliable, excluded from mean.

[‡]27 total frames, 0 speaking; excluded from mean.

*Extended head-down posture; mouth frequently occluded (see Sec. V).

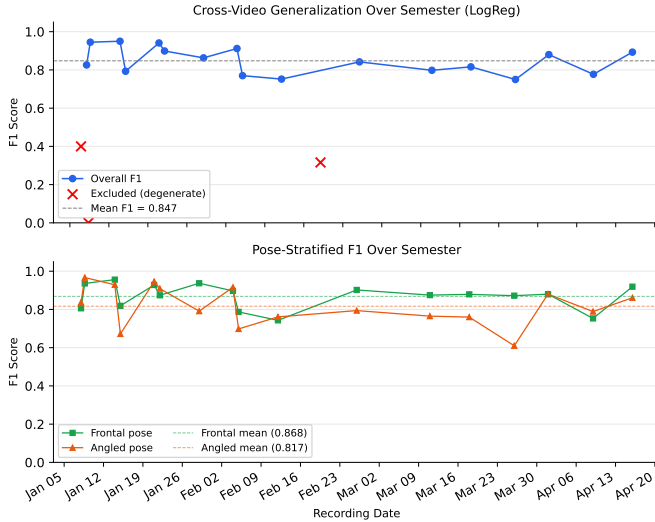


Fig. 5: Cross-video generalization over the semester (LogReg, all-features model). *Top*: overall F1 per recording; red crosses mark sessions excluded from aggregate statistics. Dashed line shows mean F1=0.848 over 17 valid recordings. *Bottom*: frontal vs. angled pose F1 over time (means 0.869 and 0.817, respectively).

ing angled F1, consistent with some degradation at larger yaw angles. However, several videos show the opposite (e.g., 20260108-1, 20260121, 20260122), suggesting that session-specific correlations between pose and speaking activity can reverse the gap. Recordings with the largest Δ_p (e.g., 20260326: 0.262, 20260219: 0.417) tend to

TABLE III: Fine-Tuning — F1 Delta vs. Zero-Shot Baseline (LogReg). $\delta_n = \text{finetuned F1} - \text{baseline F1}$ on the same held-out subset after adding n stratified target-domain frames to training.

Video	Base F1	δ_{25}	δ_{50}	δ_{100}
20260108-1	0.855	-0.038	-0.039	-0.055
20260114	0.952	-0.006	-0.020	-0.020
20260122	0.906	+0.010	+0.020	+0.023
20260204	0.830	-0.011	-0.002	+0.037
20260401	0.906	-0.019	-0.012	-0.013

involve extended off-camera or head-down postures rather than ordinary left-right yaw variation.

Motion sensitivity. High-motion vs. low-motion F1 differences average 0.042 across valid videos. Most recordings show gaps below 0.09; two exceptions (20260318: 0.119, 20260409: 0.183) have low-motion frames outperforming high-motion frames, consistent with motion blur degrading landmark quality at high frame-difference values.

Finding 2: Cross-video generalization is strong on the majority of recordings. Pose and motion sensitivity is modest on average but variable across sessions; larger gaps arise from session-specific confounds (extreme head angle, extended occlusion) rather than from systematic feature instability.

C. Experiment 3: Fine-Tuning Recovery

Table III reports the change in F1 when $n \in \{25, 50, 100\}$ stratified frames from the target video are merged with the training data and the classifier is retrained. Only 20260122 shows a consistent positive trend (+0.010 to +0.023), while 20260108-1 degrades by up to 0.055. 20260204 gains +0.037 at $n = 100$ but only after neutral results at smaller sizes, likely reflecting sampling variance with a reduced test set. The remaining two videos show negligible or slightly negative changes. No video achieves a sustained gain exceeding 0.04.

Finding 3: Adding small labeled samples from the target domain provides no consistent benefit. Because mouth openness is already a stable, geometry-based feature, the zero-shot model generalizes without adaptation for the majority of test conditions.

V. DISCUSSION AND SUMMARY

Why mouth openness dominates. Lip aperture is a direct physical correlate of voicing: producing most phonemes requires visible jaw separation. As a normalized geometric distance computed from MediaPipe landmarks, it is largely invariant to camera zoom and moderate pose changes—the landmark positions track the face, so the computed distance does not shift with head rotation within the PnP-corrected frame. This is consistent with prior work showing that lip-region features are the primary visual cue for talking-face tasks [1], [3].

Non-contribution of appearance features. Brightness, motion, and pose features each carry modest individual signal (LogReg F1 0.49–0.72), but add nothing when combined

with mouth openness. This suggests that, for a single-subject intra-session scenario, these features do not encode residual speaking-related information beyond what lip aperture already captures.

Outlier sessions. Two recordings are excluded from aggregate statistics due to degenerate label distributions. 20260108-2 contains only 27 total frames with zero speaking instances; F1 is therefore undefined (reported as 0.0). 20260107-2 contains 34 frames with only 2 speaking instances, making any F1 estimate statistically unreliable. A third session, 20260219, is retained but flagged: the subject spent an extended portion of the recording with their head down while writing, moving the mouth outside the camera’s field of view. The resulting F1 of 0.316 reflects landmark occlusion rather than feature instability under ordinary pose variation—a distinction important for interpreting the condition-stratified results.

Generalization without adaptation. Cross-video F1 (mean 0.848 on 17 valid recordings) without any domain adaptation is notable given that test recordings span a full semester and include session types with different speaking-to-silence ratios than the training video. The class-balanced classifier combined with a robust geometric feature is sufficient to bridge this distribution shift. The fine-tuning experiment (Experiment 3) confirms this: adding up to 100 target-domain frames provides no consistent improvement, indicating the zero-shot model already captures the essential signal.

Limitations. This study is *intra-subject*: all recordings feature the same speaker, so findings do not generalize to cross-subject or multi-speaker scenarios. The pipeline uses no temporal context—each frame is classified independently, which may produce jitter at speaking/silent boundaries. The corpus is CPU-processed at 3 fps, discarding the fine-grained lip dynamics that temporally dense analysis would capture. Finally, landmark occlusion (head-down posture) represents a failure mode that geometric features cannot overcome without detection fallback.

Future work. Temporal smoothing (e.g., a sliding-window majority vote or HMM post-processing) is a natural extension. Evaluating on a multi-subject corpus would test whether mouth-openness dominance holds across speakers with different articulatory styles. Audio-visual fusion could provide robustness in cases where lip occlusion prevents reliable landmark detection.

Summary. We presented a feature analysis of facial geometry for speaking detection under appearance variation. An ablation study on a single-subject lecture corpus shows that mouth openness alone matches full-feature model performance ($F1 \approx 0.883$ via 5-fold CV), and a cross-video generalization study across twenty held-out recordings confirms robust performance on seventeen evaluable sessions (mean $F1 = 0.848$). Condition-stratified analysis reveals variable pose sensitivity across sessions, with larger gaps driven by landmark occlusion rather than ordinary yaw variation. A fine-tuning experiment confirms that the zero-shot model requires no adaptation. These results argue for geometric lip aperture as a reliable,

interpretable, and computationally lightweight feature for this class of speaking-detection tasks.

REFERENCES

- [1] J. S. Chung and A. Zisserman, “Out of time: Automated lip sync in the wild,” in *Proc. ACCV Workshop on Multi-view Lip-reading*, Taipei, Taiwan, Nov. 2016.
- [2] J. S. Chung, A. Senior, O. Vinyals, and A. Zisserman, “Lip reading sentences in the wild,” in *Proc. IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR)*, Honolulu, HI, Jul. 2017, pp. 6450–6459.
- [3] J. Roth, S. Chaudhuri, O. Klejch, R. Marvin, A. Gallagher, L. Kaver, S. Ramaswamy, A. Stopczynski, C. Schmid, Z. Liu, and C. Hartmann, “Ava-activespeaker: An audio-visual dataset for active speaker detection,” in *Proc. IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR)*, Seattle, WA, Jun. 2020, pp. 5620–5629.
- [4] N. Ruiz, E. Chong, and J. M. Rehg, “Fine-grained head pose estimation without keypoints,” in *Proc. IEEE/CVF Conf. on Computer Vision and Pattern Recognition Workshops (CVPRW)*, Salt Lake City, UT, Jun. 2018.
- [5] Y. Kartynnik, A. Ablavatski, I. Grishchenko, and M. Grundmann, “Real-time facial surface geometry from monocular video on mobile GPUs,” in *Proc. IEEE/CVF Conf. on Computer Vision and Pattern Recognition Workshops (CVPRW)*, Long Beach, CA, Jun. 2019.
- [6] G. Bradski, “The OpenCV library,” *Dr. Dobbs’s Journal of Software Tools*, 2000.